

A Description of the Cleaned NHANES Dataset

Aminah Brown-Roopnarine

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Introduction

The National Health and Nutrition Examination Survey (NHANES) is a program of studies run by the National Center for Health Statistics at the Centers for Disease Control and Prevention. It combines interviews with physical examinations to assess the health and nutritional status of the non-institutionalized U.S. population (Centers for Disease Control and Prevention, National Center for Health Statistics 2024). The survey is conducted in two-year cycles, called waves, and each wave uses a nationally representative sample.

The cleaned dataset used in this report contains 29,400 participants across 3 waves. It includes demographic variables (age, gender, and ethnicity) and systolic blood pressure (SBP) measurements. This report describes the sample size and composition of each wave and summarizes average SBP over time.

Sample Size and Gender by Wave

Table 1: Overall Sample Size and Number of Males by NHANES Wave

Wave	Participants	Males	
		n	%
2015	10,175	5,003	49.2
2016	9,971	4,892	49.1
2017	9,254	4,557	49.2

Note:

Percentages are calculated out of all participants in each wave.

Table 1 shows the number of participants in each wave. Wave 2015 is the largest (10,175 participants) and wave 2017 is the smallest (9,254). The proportion of males stays fairly stable across waves, ranging from 49.1% to 49.2%, so the gender balance of the sample is consistent over time.

Ethnicity by Wave

Table 2 reports the ethnic composition of each wave. The distribution is uneven, with some groups making up a much larger share of the sample than others, and the shares change somewhat between waves. This reflects NHANES's sampling design, which oversamples certain groups to produce reliable estimates for them.

Table 2: Number and Percentage of Participants by Ethnicity and NHANES Wave

Ethnicity	Wave		
	2015	2016	2017
Asian	1,074 (10.6%)	1,042 (10.5%)	1,168 (12.6%)
Black	2,267 (22.3%)	2,129 (21.4%)	2,115 (22.9%)
Mexican	1,730 (17.0%)	1,921 (19.3%)	1,367 (14.8%)
Other	470 (4.6%)	505 (5.1%)	634 (6.9%)
Other Hispanic	960 (9.4%)	1,308 (13.1%)	820 (8.9%)
White	3,674 (36.1%)	3,066 (30.7%)	3,150 (34.0%)

Note: Cells show n (column %). Percentages exclude participants with missing ethnicity.

Systolic Blood Pressure by Wave

Table 3: Summary Statistics of Average Systolic Blood Pressure (mm Hg) by NHANES Wave

Wave	Average SBP (mm Hg)				
	Min	Max	Median	Mean	SD
2015	64.7	228.7	115.3	118.3	18.1
2016	74.0	231.3	117.3	120.4	18.4
2017	72.7	234.0	118.0	121.7	20.3

Note:
Average SBP is the mean of up to four readings per participant. The highlighted row marks the wave with the highest mean.

Table 3 summarizes average SBP by wave. Mean SBP ranges from 118.3 to 121.7 mm Hg, and the highest mean occurs in wave 2017 (highlighted). Medians are close to the means, which suggests the distributions are only mildly skewed, and the standard deviations are similar across waves, so the spread of blood pressure is stable over time.

Conclusion

The cleaned NHANES dataset provides a large, consistent sample across waves, with a stable gender balance, an uneven but well-documented ethnic composition, and average systolic blood pressure that changes only modestly from one wave to the next.

References

Centers for Disease Control and Prevention, National Center for Health Statistics. 2024. *National Health and Nutrition Examination Survey Data*. U.S. Department of Health; Human Services. <https://www.cdc.gov/nchs/nhanes/>.